Amortizing Force-Aware Retargeting for Scalable Human-to-Robot Data Generation

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Project Page

Motivation & Task

Amortize = Reduce repeated computation by learning reusable corrections upfront.

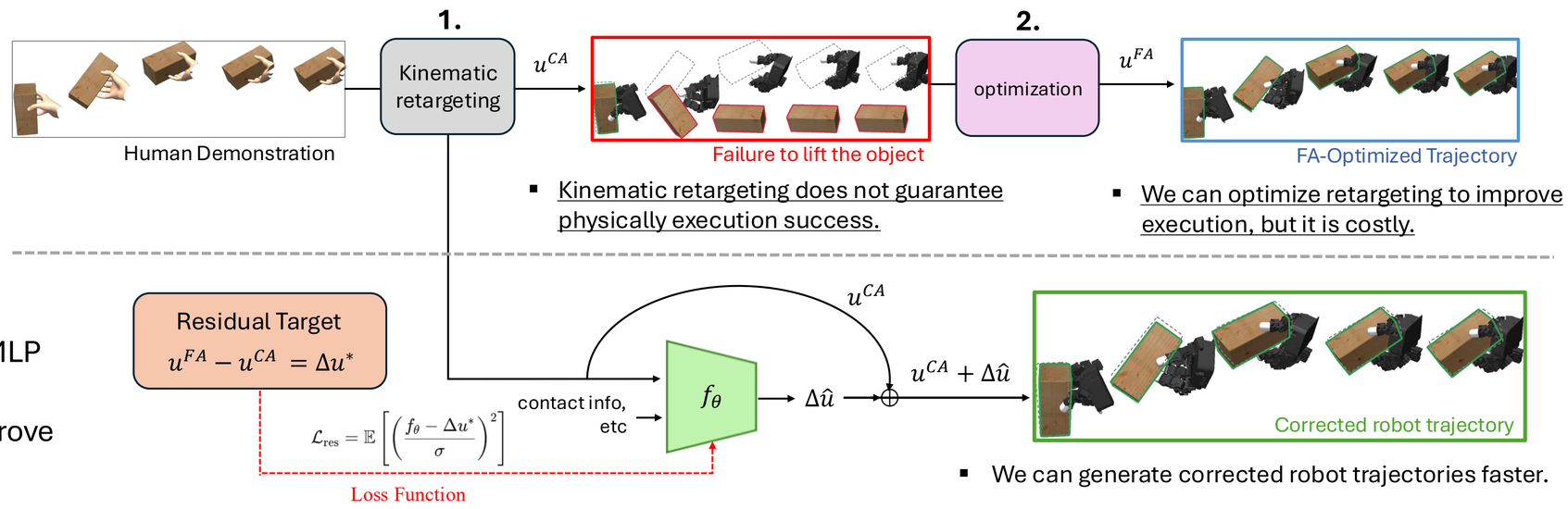
We learn control corrections from DexForge [1] to reduce the cost of converting human demonstrations into robot trajectories.

[1] M. Wang, K. Cao, R. Ni, L. Xie, and Y. Hong, "DexForge: High-Fidelity Physics-Informed Dexterous Retargeting," *arXiv preprint arXiv:2610.06331*, 2026.

[2] Y.-W. Chao et al., "DexYCB: A Benchmark for Capturing Hand Grasping of Objects," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.

Previous Method

1. Contact-Aware Retargeting: Adapts human contact points to the robot hand and generates a reference joint trajectory.
2. Force-Aware Retargeting: Refines the CA reference by optimizing position-control commands through simulated contact forces and dynamics to track the target object motion.



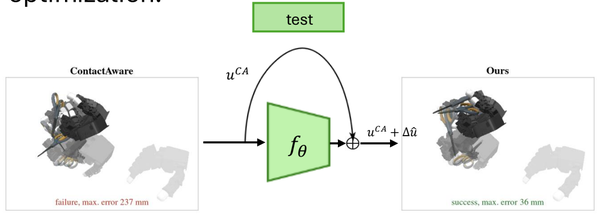
Proposed Method: Train a lightweight MLP to predict residual corrections and add them to kinematic retargeting reference controls to improve physical execution success.

Data Generation & Application

- **Before Training:** Generate paired CA–FA controls from DexYCB [2] demonstrations to train a residual MLP.

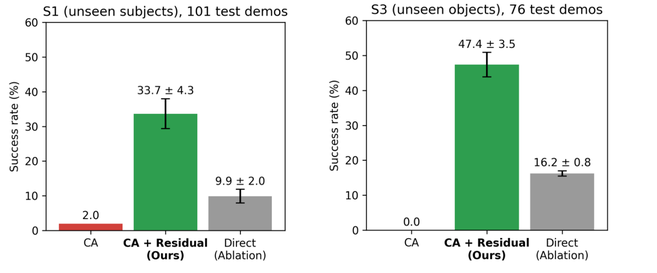


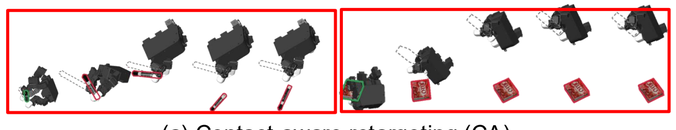
- **After Training:** Add predicted residuals to CA controls for unseen demonstrations, avoiding repeated optimization.



Evaluation Results

Our residual corrections increase success from **2.0% to 33.7% on S1** and from **0.0% to 47.4% on S3**.





(a) Contact-aware retargeting (CA)

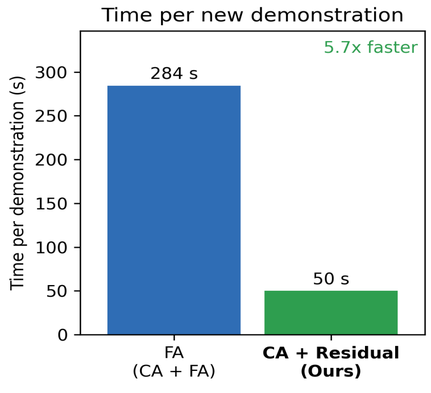


(b) CA+Residual (Ours)

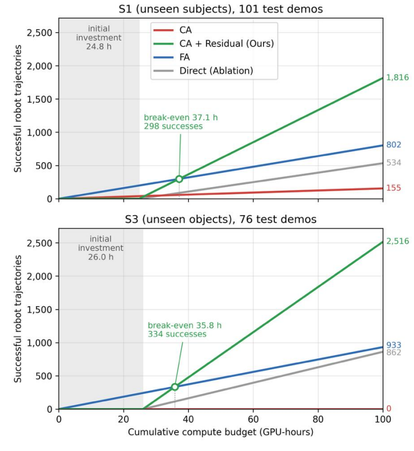
More Demos !!
namaewa-im.github.io/amortizing_retargeting/

Discussion

1. Lower computation time for new input demonstration. 5.7× faster



2. Break-Even Compute Time Budget
37.1 GPU-hours (S1), 35.8 GPU-hours (S3).



Future work

Scale up robot training data generation by retargeting human demonstrations from web videos into physically executable robot motions.

Buy the dip! Connect now!